Course Title: Database Management System Lab

Course Code: CSE 224

Project Presentation on Movie Ticket Booking System

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Movie Ticket Booking System

1. Introduction

The Movie Ticket Booking System is a web-based application designed to simplify the process of booking movie tickets online. The system allows users to browse movies, check showtimes, select seats, and book tickets conveniently.

This system integrates frontend, backend, and database technologies to create a complete online booking platform. Users will be able to register, log in, view available movies, and book tickets, while administrators will manage movies, show schedules, and user accounts.

The main focus of this project is to demonstrate database design, CRUD operations, user authentication, and an interactive user interface.

2. Project Objective

The main objective of this project is to design and develop a web application that integrates database systems with frontend and backend technologies.

Specific objectives include:

- Implementing full CRUD operations (Create, Read, Update, Delete)
- Designing a structured MySQL database
- Creating a user-friendly interface
- Implementing user authentication and authorization

• Managing movies, showtimes, and ticket bookings efficiently

3. Technology Stack

The system will be developed using the following technologies**.**

Frontend:

•HTML

• CSS

• JavaScript

Backend :

Node.js

Express

Database

MySQL/PHPmyadmin (via xampp)

These technologies will work together to handle user interaction, server procession, and data storage.

4. System Architecture

The application will follow a three-layer architecture.

1. Presentation Layer

The frontend interface built using React allows users to interact with the system through movie listings, booking forms, and dashboards.

***Responsive design:** Mobile Friendly Design for seamless access on various screen sizes.

2. Application Layer

The backend developed with Node.js and Express.js processes user requests, manages ticket booking logic, and handles authentication.

3. Data Layer

The MySQL database stores all system information including users, movies, showtimes, bookings, and seats.

4. System Modules

The system will include two main modules:

• User Panel

• Admin Panel

5. User Panel Features

The User Panel allows general users to interact with the system.

1. User Registration and Login

Users can create accounts and log in securely.

Functions include:

• User registration

• Secure login authentication

• Logout functionality

2. User Profile Management Users can manage their personal information.

Features include:

• View profile

• Update personal information

• Change password

3. Movie Browsing

Users can browse available movies in the system.

Features include:

• View movie list

• View movie details

• View showtimes

•Ticket Booking

• Select seats

•Check booking summary

• Confirm ticket booking

4. Booking History

Users can cancel their bookings.

Functions include:

• Select movie

• Choose showtime

• Select seats

• Confirm ticket booking

5. Booking History

Users can view their previous bookings.

Features include:

• View booked tickets

• Booking details

• Booking date and time

6. Search and Filter

Users can easily find movies.

Features include:

• Search movie by title

• Filter by genre

• Sort movies by release date

6. Admin Panel Features

The Admin Panel allows administrators to manage the entire system.

1.Admin panel Features

The dashboard provides an overview of system statistics.

Information includes:

• Total users

• Total movies

• Total bookings

2. Movie Management

Admins can manage movies in the system. Functions include:

• Add new movies

• Edit movie information

• Delete movies

3. Show Schedule Management

Admins can manage movie showtimes.

Features include:

• Add show schedule

• Update showtime

• Delete schedule

4. User Management

Admins can manage user accounts.

Functions include:

• View all users

• Delete user accounts

7. Database Design

The system database will consist of several tables.

Users Table

• user_id (Primary Key)

• name

• email

• password

• role

Movies Table

• movie_id (Primary Key) • movie_title

• genre

• duration

• release_date

• description

Showtimes Table

• showtime_id (Primary Key)

• movie_id (Foreign Key)

• show_date

• show_time

•Hall_number

Bookings Table

• booking_id (Primary Key)

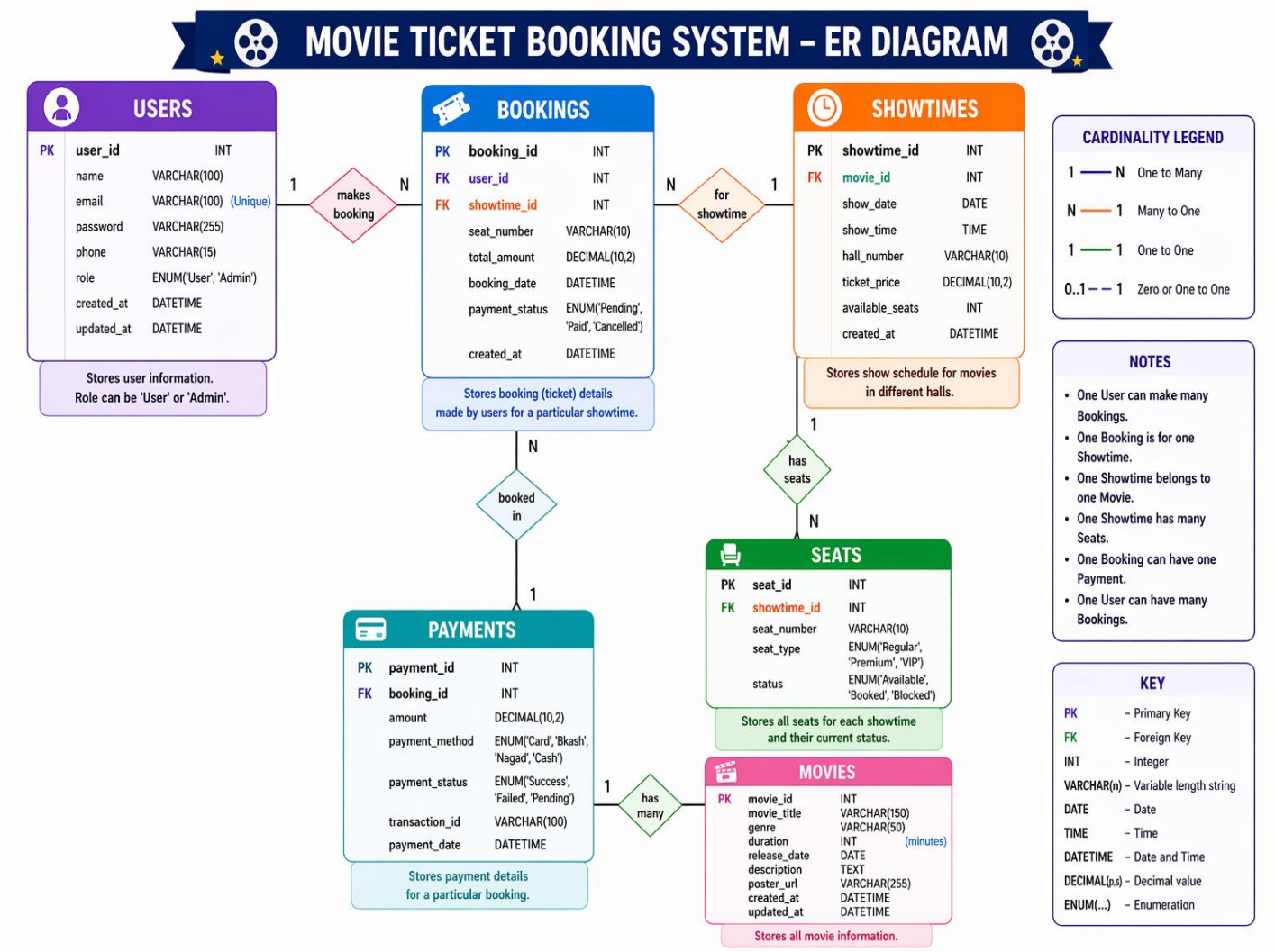
• user_id (Foreign Key)

• showtime_id (Foreign Key)

• seat_number

• booking_date

8. ER Diagram



9. System Workflow

The system workflow explains how the application processes user actions. 1. The user accesses the system through the frontend interface.
2. The user registers or logs in to the system.

3. The user browses available movies and showtimes.

4. The user selects seats and books tickets.

5. The frontend sends requests to the backend server.

6. The backend processes the request and communicates with the MySQL database.

7. The database returns the requested data.

8. The backend sends the response to the frontend.

9. The booking confirmation is displayed to the user.

10. Security Measures

To ensure system security, the following techniques will be implemented:

• User authentication system

• Password encryption

• Input validation

• Secure database queries

• Role-based access control (Admin/User)

11. Expected Outcome

The final system will provide a complete platform for online movie ticket booking. The project will demonstrate:

• Full-stack web application development

- Database integration
- CRUD operations
- User authentication
- Interactive user interface

This system will serve as a practical example of implementing a modern web application using frontend, backend, and database technologies.

Contribution List

1.Nasruma Hossain Nuha ID:241-115-256

Contributions: (Frontend)

Contributions:

• Designed and developed the frontend of the Movie Booking System using HTML and CSS for a clean and responsive user interface.

• Created the layout and structure of web pages, including homepage, movie listings, booking interface, and authentication pages.

• Applied consistent styling, color themes, and responsive design techniques to improve user experience across different devices.

• Designed UI components such as movie cards, search and filter layout, seat selection interface, and booking summary section.

• Designed seat booking visuals with clear indicators for available, selected, and booked seats.

• Ensured smooth navigation and user-friendly interface for browsing movies and completing bookings.

• Tested and refined the interface to ensure proper display and usability.

2.Jannatul Millati ID: 241-115-017

Contributions: (Backend)

•Designed and developed the backend system using Node.js and Express.js for the Movie Booking System.

•Configured and connected the backend with MySQL database (XAMPP) to manage movie data efficiently.

•Created and structured the database movie_booking with the movies table to store movie title and genre.

•Implemented RESTful APIs (e.g., GET /movies) to fetch movie data and send responses in JSON format.

•Managed environment configuration using .env file for secure database credentials and server settings.

•Tested and debugged backend functionalities, ensuring proper server execution and database connectivity.

•Integrated backend APIs with the frontend to display movie data dynamically on the website.

2.Rima Rimjhim Chisim ID:241-115-018

Contributions: (Functionalities & ER Diagram)

• Defined all core functionalities of the system

• Worked on user flow (registration → login → booking → payment)

• Designed ER Diagram with proper entities and relationships

• Identified entities like Users, Movies, Booking, Payment, Showtimes

• Maintained relationship mapping (one-to-many, many-to-many)

• Ensured proper connection between functionalities and database

• Analyzed system requirements before implementation

• Helped in planning overall system workflow

• Supported integration of frontend, backend, and database

• Assisted team in understanding project structure

4.Prianka Halim Samia. ID:241-115-011

Contributions: (Database Design)

* Designed and developed the MySQL database for a Movie Booking System, including users, movies, showtimes, seats, and bookings tables.

* Established relationships between tables using foreign keys to ensure proper data integrity and structure.

* Implemented SQL queries for data insertion and automated seat generation for all showtimes.

* Managed booking logic support through seat availability tracking and user role-based structure.